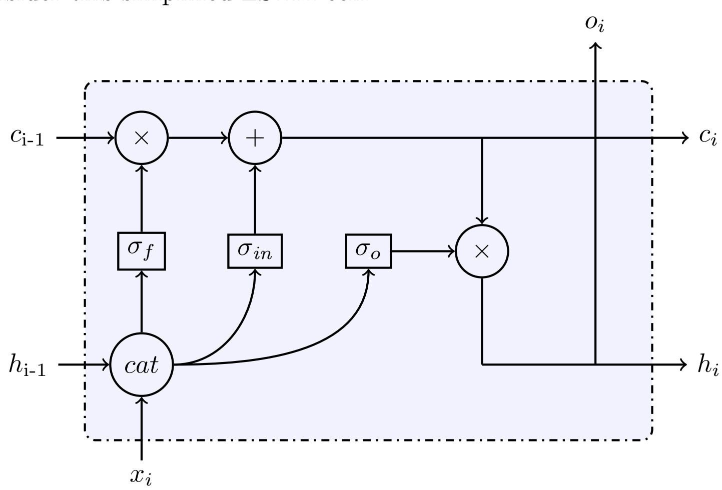
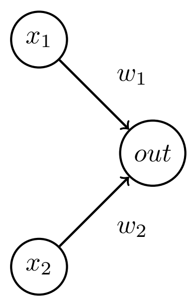
1 LSTM

Consider this simplified LSTM cell:



Assume all three gates (σ_x) to be simple feed-forward networks with two inputs and one output:



Assume the Bias to be 0. *Cat* stands for the concatenation operation. **Example:** $x_1 = 5, x_2 = 3 : cat(x_1, x_2) \rightarrow [1, 5]$.
$\times$ and $+$ are multiplication and addition. Assume the following activation function for all σ:

$$f(x) = \left\{ \begin{array}{ll} 2, & \text{if } x \geq 2 \\ 1, & \text{if } x < 2 \end{array} \right\}$$

Given the following weights and inputs, calculate the values for o_i, c_i and h_i!

INPUTS:

$h_{i\text{-}1} = 2$

$c_{i\text{-}1} = 1$

$x_i = 3$

WEIGHTS:

$w_{f1} = 0$, $w_{f2} = 0.6$

$w_{in1} = 0.5$, $w_{in2} = 0.5$

$w_{o1} = 0.25$, $w_{o2} = 0.5$